INTELLIGENT ENERGY METER WITH REAL TIME MONITORING

A PROJECTREPORT

of

ECB1303 – EMBEDDED SYSTEM AND IOT DESIGN

Submitted by

NITHYA SHRREE S (8115U24EC072)
SAMEEHA FATHIMA S (8115U24EC093)
SUBASRRI D (8115U24EC110)

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING
in

ELECTRONICS AND COMMUNICATION ENGINEERING

Under the Guidance of

Dr. R. SAMSON DANIEL,M.E.,(Ph.D)

Associate Professor/ECE

Department of Electronics and Communication Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
Under
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BONAFIDE CERTIFICATE

Certified that this project report titled "**INTELLIGENT ENERGY METER WITH REAL TIME MONITORING**" is the bonafide work of **NITHYA SHRREE S (8115U24EC072), SAMEEHA FATHIMA S (8115U24EC093), SUBASRRI D (8115U24EC110)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Submitted for the End Semester Examination held on …….

INTERNAL EXAMINAR **EXTERNAL EXAMINAR**

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ANNA UNIVERSITY, CHENNAI

DECLARATION BY THE CANDIDATES

We declare that to the best of our knowledge the work reported herein has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voce held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATES

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DEPARTMENT OFECE

VISION

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION

M1. To equip the students with latest technical, analytical and practical knowledge

M2. To provide vibrant academic environment and Innovative Research culture

M3. To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

PEO1: Graduates will become experts in providing solution for the Engineering problems in Industries, Government and other organizations where they are employed.

PEO2: Graduates will provide innovative ideas and management skills to enhance the standards of the society by individual and with team works through the acquired Engineering knowledge.

PEO3: Graduates will be successful professionals through lifelong learning and contribute to the society technically and professionally.

PROGRAMSPECIFIC OUTCOMES (PSO'S):

PSO1: Analyze and design complex electronic circuits and communication systems by applying the principles of engineering and mathematics.

PSO2: Analyze and provide innovative design solutions to research problems, industrial problems and for the demanding needs of the society in the domains of electronic integrated circuits, signal processing, high frequency and antenna design.

PSO3: Continuously involve in life-long learning activities so as to evolve and become an active team member in diverse and challenging design situations.

PROGRAM OUTCOMES:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and the World: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PO8: Individual and Collaborative Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11: Lifelong learning: Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

CO-PO MAPPING FOR
EMBEDDED SYSTEMS AND IOT DESIGN

After successful completion of this course, the students should be able to

CO1: Analyze the internal architecture and modelling of an embedded processor using design tools.

CO2: Develop the embedded system networks using different protocols.

CO3: Identify the evolution of the Internet of Things (IoT) using internal components for networking.

CO4: Evaluate cloud and big data analytics using IoT Platform.

	POs											PSOs			
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Average
CO1	2.60	1.40	2.00	1.80	1.80	3.00	1.40	1.80	1.40	2.00	3.00	1.80	1.40	0.80	1.87
CO2	2.60	1.20	2.00	1.20	2.00	3.00	1.20	1.40	1.60	1.80	3.00	1.80	1.80	0.20	1.77
CO3	2.60	1.20	2.00	1.60	1.00	3.00	1.20	1.60	1.80	1.20	3.00	1.80	1.80	0.80	1.76
CO4	2.60	1.20	1.20	2.00	1.20	3.00	1.20	1.80	1.60	1.20	3.00	1.80	1.80	1.20	1.77
CO5	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
Average	2.68	1.60	2.04	1.92	1.80	3.00	1.60	1.92	1.88	1.84	3.00	2.04	1.96	1.20	2.03

identify and select the most appropriate Sustainable Development Goals (SDGs) that align with the objectives and outcomes of the project.

Sustainable Development Goals (SDGs)

☐ **Goal 1: No Poverty** – End poverty in all forms everywhere

☐ **Goal 2: Zero Hunger** – End hunger and improve nutrition

☐ **Goal 3: Good Health and Well-being** – Ensure healthy lives

☐ **Goal 4: Quality Education** – Provide inclusive, equitable education

☐ **Goal 5: Gender Equality** – Empower women and girls

☐ **Goal 6: Clean Water and Sanitation** – Ensure safe water and sanitation

☐ **Goal 7: Affordable and Clean Energy** – Access to sustainable energy

☐ **Goal 8: Decent Work and Economic Growth** – Promote jobs and growth

☐ **Goal 9: Industry, Innovation and Infrastructure** – Build resilient infrastructure

☐ **Goal 10: Reduced Inequalities** – Reduce inequality

☐ **Goal 11: Sustainable Cities and Communities** – Make cities safe and sustainable

☐ **Goal 12: Responsible Consumption and Production** – Reduce waste

☐ **Goal 13: Climate Action** – Combat climate change

☐ **Goal 14: Life Below Water** – Protect oceans

☐ **Goal 15: Life on Land** – Protect ecosystems and biodiversity

☐ **Goal 16: Peace, Justice and Strong Institutions** – Promote justice and peace

☐ **Goal 17: Partnerships for the Goals** – Strengthen global partnerships

ACKNOWLEDGEMENT

It is with great pride that we express our gratitude and indebtedness to our institution, "**K. Ramakrishnan College of Engineering (Autonomous)**", for providing us with the opportunity to do this project.

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We thank **Dr. M. MAHESWARI, M.Tech., Ph.D.,** Head of the Department of **ELECTRONICS AND COMMUNICATION ENGINEERING**, for providing her encouragement in pursuing this project.

We wish to convey our profound and heartfelt gratitude to our esteemed project guide and Course Coordinator **Dr. R. SAMSON DANIEL M.E., (Ph.D),** Associate Professor, Department of **ELECTRONICS AND COMMUNICATION ENGINEERING,** for his incalculable suggestions, creativity, assistance and patience, which motivated us to carry out this project.

We wish to express our special thanks to the officials and Lab Technicians of our departments who rendered their help during the period of the work progress.

ABSTRACT

This project presents the design and implementation of an IoT-Based Intelligent Energy Management and Abnormal Load Monitoring System to address the critical limitations of conventional electricity metering. Traditional meters calculate usage only at the end of a billing cycle, offering no proactive warnings before consumers cross into expensive tiered electricity tariff slabs, and they lack the capability to detect hidden energy waste caused by degrading or faulty appliances.

To overcome these issues, the proposed system integrates total household energy metering with real-time, branch-level appliance monitoring. A commercial single-phase digital energy meter is interfaced with an Arduino Uno microcontroller via a PC817 optocoupler to safely track cumulative power consumption. The system is programmed to issue automated IoT alerts at critical billing milestones—specifically at 80, 450, and 950 units—to help users manage consumption before incurring higher slab charges.

Simultaneously, a dedicated Current Transformer (CT) sensor continuously monitors individual load lines to identify abnormal current spikes indicative of appliance faults, such as failing motors or compressors. Real-time electrical parameters are displayed locally on a 16x2 LCD and transmitted to a cloud dashboard via a NodeMCU (ESP8266) Wi-Fi module for remote user monitoring.

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

Traditional electricity meters leave consumers vulnerable to expensive tiered tariff penalties and hidden fire hazards by providing only aggregated, monthly consumption data that obscures real-time usage and individual appliance inefficiency. To address this, the "IoT-Based Intelligent Energy Management and Abnormal Load Monitoring System" combines real-time main-line smart metering with branch-level current sensing.

The system continuously calculates cumulative power consumption and sends dynamic IoT alerts at critical tariff milestones (80, 450, and 950 units) to help users proactively avoid billing spikes. Additionally, by utilizing dedicated Current Transformer (CT) sensors on individual branch lines, the system monitors specific appliances for abnormal current spikes or persistent overloads caused by degrading components.

Upon detecting a fault, it provides instant internet-based diagnostics and automatically triggers a relay to disconnect the hazardous load, ultimately empowering users to optimize their energy costs while significantly improving household electrical safety.

1.2 TECHNOLOGIES USED.

- **Optocoupler Isolation Technology:** Uses a PC817 optocoupler to provide galvanic isolation, converting high-voltage optical pulses into safe 5V digital signals.
- **Current Transformer (CT) Technology:** Employs CT sensors (like CT1270) non-invasively clamped onto branch lines to measure live analog alternating current drawn by individual appliances.
- **Microcontroller Technology:** Uses an Arduino Uno for centralized data acquisition, interrupt handling, tariff calculation, and load control.
- **IoT/Wi-Fi Technology:** Utilizes a NodeMCU (ESP8266) to transmit telemetry data to cloud dashboards for remote monitoring and mobile alerts.
- **Relay Actuation:** Uses a 5V driver relay module to automatically disconnect faulty loads during overcurrent events.
- **Embedded C/C++ Programming:** Used to program the microcontrollers for hardware interrupts, ADC conversions, and serial communications.
- **LCD Display Technology:** A 16x2 character LCD provides localized, real-time readouts of energy metrics and system status.

1.3 MICROCONTROLLER – ESP32

The proposed system employs a dual-microcontroller architecture to separate core processing from wireless networking tasks.

The **Arduino Uno** serves as the central computational and hardware interfacing engine of the entire system. It is strictly responsible for capturing high-speed hardware interrupts generated by the energy meter's optical pulse output, which is safely routed through the PC817 optocoupler. Simultaneously, the Arduino's built-in Analog-to-Digital Converter (ADC) continuously samples the conditioned analog voltage signals received from the branch-level Current Transformer (CT) sensors. Using this raw data, the Arduino executes the core decision-making algorithms: it calculates the cumulative kilowatt-hour (kWh) usage, computes the true Root Mean Square (RMS) current of individual appliances, and evaluates this real-time data against dynamic tiered tariff slabs and overcurrent safety thresholds. Upon processing these parameters, the Arduino directly updates the local 16x2 character LCD and sends instant digital control signals to the 5V relay driver module to physically isolate any malfunctioning loads.

Operating alongside the Arduino is the NodeMCU (ESP8266), which functions as a dedicated Wi-Fi transceiver. Because the Arduino handles the intensive real-time sensing and pulse-counting, it simply packages the processed electrical parameters into structured data packets and transmits them to the ESP8266 via UART serial communication.

The NodeMCU is programmed with network-specific firmware to establish a local Wi-Fi connection and utilizes web protocols to push this incoming telemetry to a cloud-based IoT dashboard. This division of labor ensures highly stable,

uninterrupted power calculation by the Arduino, while the ESP8266 seamlessly manages the remote user interface, cloud data visualization, and automated mobile push notifications for tariff warnings and fault alerts

OBJECTIVE

The primary objective of this project is to develop an IoT-Based Intelligent Energy Management and Abnormal Load Monitoring System that resolves the financial and operational inefficiencies of current electricity monitoring. The system aims to provide a dynamic slab warning mechanism to track live power consumption and issue automated IoT alerts at vital billing milestones, empowering users to avoid tiered tariff penalties.

Furthermore, it aims to monitor branch-level appliance currents to detect hidden energy waste, provide automated electrical fault diagnostics, and safely disconnect specific malfunctioning loads without shutting down the main power supply.

1.4 FEATURES

The The IoT-Based Intelligent Energy Management System continuously monitors household electricity consumption by integrating a commercial energy meter with an Arduino Uno. By safely capturing optical calibration pulses, the system accurately calculates real-time kilowatt-hour (kWh) usage, eliminating the reliance on delayed, end-of-month utility bills.

A key feature is the dynamic tiered tariff alert mechanism, designed to help consumers avoid steep billing penalties. The system actively tracks power consumption and sends automated IoT push notifications to a mobile device or cloud dashboard when usage reaches critical milestones of 80, 450, and 950 units.

To overcome the limits of aggregated whole-house metering, dedicated Current Transformer (CT) sensors monitor individual branch lines for real-time fault diagnostics. The system detects abnormal current spikes from degrading appliances and automatically triggers a relay to isolate the faulty device without disrupting household power. Additionally, it displays live metrics locally on a 16x2 LCD while transmitting data via a NodeMCU (ESP8266) Wi-Fi module for remote cloud monitoring.

CHAPTER 2

LITERATURE SURVEY

i. **IoT Based Smart Energy Meter Monitoring and Billing System – Manisha Tejwani, Prachi Rane, Moin Syed, Oyesh Patel, Pragya Pandey – 2020**

Key Findings:

- Uses ATmega328P and ESP8266 for real-time pulse counting and ThingSpeak billing reports.

Disadvantages:

- It only tracks total mains consumption.
- Lacks threshold warnings or appliance-level monitoring.

ii. **IoT Based Energy Meter Billing System – P. S. Vaidya, C. S. Kature, S. G. Waghmare, S. A. Dhumane – 2018**

Key Findings:

- Uses Arduino and Wi-Fi to send automated billing updates and basic budget cap alerts via SMS/web.

Disadvantages:

- It uses static percentage caps instead of utility tariff slabs.

- Lacks hardware to isolate individual appliance consumption.

iii. **IoT based Energy Meter with Smart Monitoring of Home Appliances – Vishnukant V. Gavhane, Mayuri R. Kshirsagar, Ganesh M. Kale, Shubham Katangle – 2021**

Key Findings:

- Combines main-line billing (ZMPT101B/ACS712) with remote relay switching for home appliances.

Disadvantages:

- Appliance tracking is limited to ON/OFF relay states rather than measuring live individual current consumption.

iv. **Design and Implementation of an Intelligent Smart Energy Meter Using IoT and Machine Learning – R. Sharma, A. Kumar, P. Verma, S. Singh – 2024**

Key Findings:

- Integrates ESP32 with sensors and a Machine Learning model to detect power theft and total load anomalies.

Disadvantages:

- Anomaly detection is restricted to the main supply line.
- Cannot identify which specific appliance is faulty.

v. **Smart Energy Meter Using IoT and Offline Data Storage – M. A. Patel, S. N. Joshi, V. K. Patel – 2022**

Key Findings:

- Uses NodeMCU with ACS712 and ZMPT101B for RMS power calculation and grid over/under-voltage protection.

Disadvantages:

- Focuses strictly on main-line protection rather than appliance fault diagnosis or tariff slab alerts.

vi. **IoT-Based Smart Energy Meter with Real-Time Fault and Anomaly Analytics – K. R. Suresh, B. Venkatesh, D. Anitha – 2023**

Key Findings:

- Employs an ESP32 and cloud dashboard (Firebase/Blynk) for high-sampling RMS monitoring and global overload relay trips.

Disadvantages:

- Fault detection operates globally on the main line using fixed cut-offs instead of tiered slab alerts.

vii. **IoT Based Smart Energy Meter using PZEM-004T and NodeMCU – M. R. Singh et al. – 2022**

Key Findings:

- Utilizes a PZEM-004T module and NodeMCU ESP32 for real-time energy monitoring via
  cloud-based Firebase storage.
- Provides a user-friendly interface via the Blynk application to track voltage, current, and power trends.

Disadvantages:

- Cannot isolate power consumption for individual branch circuits or detect specific appliance faults.
- Focuses on total load monitoring without automated tiered tariff slab alerts.

viii. **Smart Energy Metering and Load Controlling by Using Internet of Things – A. Kumar et al.**
– 2022

Key Findings:

- Implements a NODE MCU and ACS712 current sensor to transmit energy usage data to Wi-Fi server.
- Automates output loads using a relay module controlled via a smartphone

application.

Disadvantages:

- Accuracy of ACS712 sensors degrades with electromagnetic noise compared to commercial digital energy meters and CT sensors.
- Does not provide proactive billing alerts at specific multi-tier tariff milestones.

ix. **IoT Smart Electricity Energy Meter for Household Appliance Monitoring – S. Gupta et al. – 2024**

Key Findings:

- Proposes an IoT system for real-time insights into energy consumption at both the household and device levels.
- Enhances transparency by calculating and displaying energy costs to help users optimize electricity expenses.

Disadvantages:

- High implementation cost and wiring complexity for installing sensors directly on every household device.
- Lacks an automated physical disconnection relay for immediate overload and short-circuit protection.

x. **Development of an IoT-Enabled Smart Electricity Meter– H. O.Garcés et al. – 2025**

Key Findings:

- Develops an advanced Smart Energy Meter (SEM) that transmits high-speed realtime data via IoT.
- Achieves superior accuracy in measuring complex electrical parameters, including line harmonics and frequency.

Disadvantages:

- Highly complex data processing requirements that exceed the capabilities of standard microcontrollers like the Arduino Uno.
- Over-engineered and too expensive for simple residential fault detection and tiered billing management.

xi. **IoT-Based Smart Energy Meter System with Home Automation P. Sharma et al. – 2024**

Key Findings:

- Integrates energy monitoring with IFTTT voice commands and MQTT protocols using a NodeMCU microcontroller.
- Converts traditional home appliances into smart devices accessible remotely via mobile applications.

Disadvantages:

- Focuses heavily on smart home automation (like motion sensors) rather than dedicated electrical fault diagnostics.
- Intermittent Wi-Fi stability issues can interrupt the continuous pulse counting

required for accurate power billing.

xii. **IOT Based Fault Detection and Protection of Power Transformer in the Smart Grid – M.Ali et al. – 2021**

Key Findings:

- Replaces traditional differential relays with an Arduino-based monitoring system for real-time fault detection.
- Senses current, voltage, and temperature autonomously to generate tripping signals for line-to-ground and unsymmetrical faults.

Disadvantages:

- Designed specifically for large-scale grid power transformers rather than domestic household appliances.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

The evolution of energy monitoring has shifted toward smart solutions, yet most systems still focus exclusively on main-line billing rather than comprehensive energy management. Standard meters measure only total household consumption, treating all connected appliances as a single aggregated load and making individual monitoring impossible. Furthermore, they operate on a reactive monthly billing cycle, failing to provide proactive warnings when users approach expensive multi-tier tariff limits.

Early smart meters successfully digitized the billing process for remote reading but remained limited in functionality. They only track total mains consumption and lack dynamic threshold warnings or appliance-level capabilities. While some IoT billing systems offer basic budget alerts, these typically rely on static percentage caps rather than aligning with realistic, multi-tier utility tariff slabs.

Recent advancements have attempted to integrate appliance control and analytics, but they lack the necessary granularity. Existing appliance tracking is typically restricted to recognizing the ON/OFF states of control relays rather than measuring the live, continuous current of individual devices. Even when

advanced anomaly detection is used, it operates globally on the main supply line, failing to achieve component-level isolation to identify exactly which specific appliance is faulty or degrading

3.2 BLOCK DIAGRAM OF THE EXISTING SYSTEM

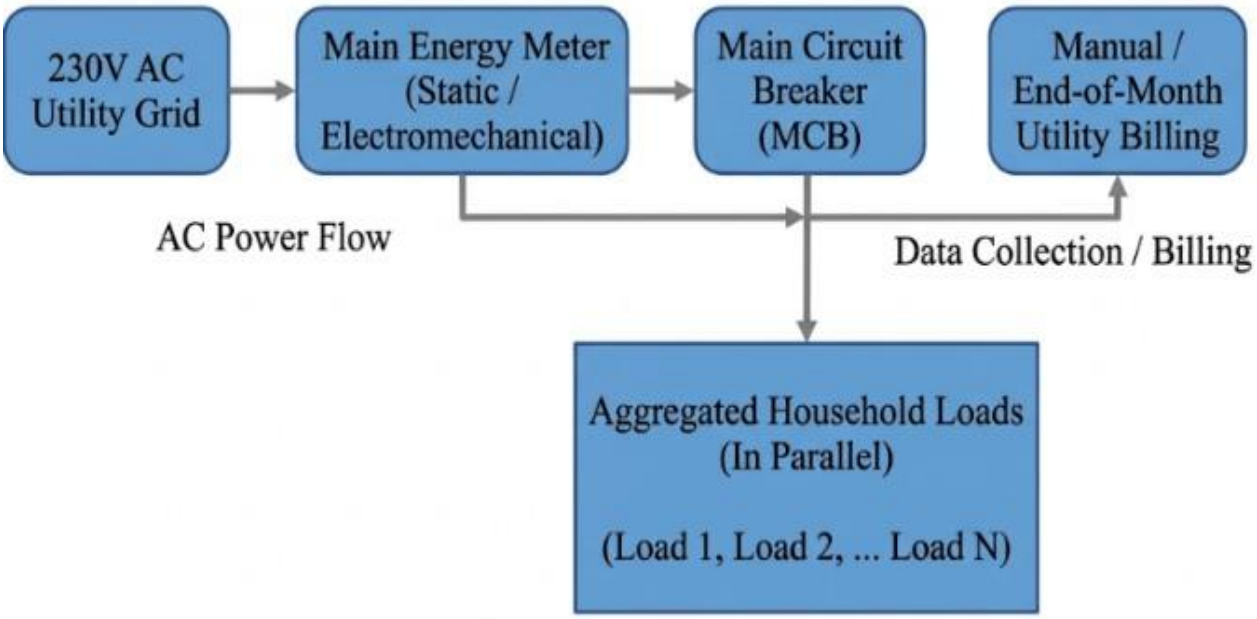


Fig:3.2.1 BLOCK DIAGRAM

The **Fig:3.2.1** shows that,

- **230V AC Utility Grid:** This is the primary external power source supplied by the regional electrical distribution network. It delivers standard high-voltage alternating current directly to the household premises.
- **Main Energy Meter (Static / Electromechanical)**: This device acts as the single point of measurement for all incoming electrical power used by the household. It continuously aggregates total consumption without differentiating between individual appliances.

- **Main Circuit Breaker (MCB)** This serves as the primary global safety switch that protects the entire house from severe short circuits or massive overloads. However, it is incapable of detecting minor overcurrents caused by individual degrading appliances.
- **Aggregated Household Loads (In Parallel)** All electrical appliances and fixtures in the house are connected to this single distribution line. Because they are wired in parallel downstream of the main meter, their individual power usage cannot be isolated or monitored.
- **Manual / End-of-Month Utility Billing** This represents the reactive data collection process where meter readings are taken manually at the end of a billing cycle. It prevents users from seeing real-time usage data and offers no proactive warnings for tiered tariff limits.

3.3 PROBLEM STATEMENT

Conventional smart meters measure only total household consumption at the main supply line, relying on delayed, end-of-month billing cycles. This reactive approach leaves consumers unaware of real-time usage, frequently causing them to cross into expensive multi-tier tariff slabs without any proactive warnings or financial protection.

Furthermore, existing systems treat all connected appliances as a single aggregated load, making individual monitoring impossible. When aging appliances degrade and draw continuous excess current, standard main circuit

breakers often fail to trip. These minor but persistent overcurrents go unnoticed, leading to hidden energy waste and potential electrical fire hazards over time.

Even recent IoT solutions focus primarily on main-line anomaly detection or superficial ON/OFF appliance tracking. Consumers currently lack systems with granular branch-level current measurement and automated fault diagnostics necessary to safely isolate specific malfunctioning devices and proactively manage their energy footprint.

CHAPTER 4

PROPOSED SYSTEM

4.1 PROJECT DESCRIPTION

The proposed system introduces a dual-layered, IoT-enabled intelligent energy management framework designed to overcome the limitations of conventional whole-house electricity metering. It integrates main-line cumulative power billing with real-time, branch-level appliance monitoring to provide proactive tariff warnings and automated fault protection. By tapping into the calibration pulse of a commercial single-phase energy meter, the system continuously calculates cumulative household power consumption and triggers automated IoT warnings when usage reaches critical billing milestones of 80, 450, and 950 units to prevent bill shock.

The physical architecture is built around an Arduino Uno microcontroller acting as the central processing unit, working in tandem with a NodeMCU (ESP8266) dedicated to Wi-Fi and IoT cloud communication. To overcome whole-house aggregation, the system employs dedicated Current Transformer (CT) sensors on individual appliance branch lines to continuously monitor live current consumption. If an abnormal current spike or persistent overload is detected, the microcontroller provides instant electrical fault diagnostics to the cloud dashboard and automatically triggers a relay module to safely isolate the

faulty load, while simultaneously displaying all real-time telemetry on a local 16x2 LCD.

4.2 BLOCK DIAGRAM OF THE PROPOSED SYSTEM

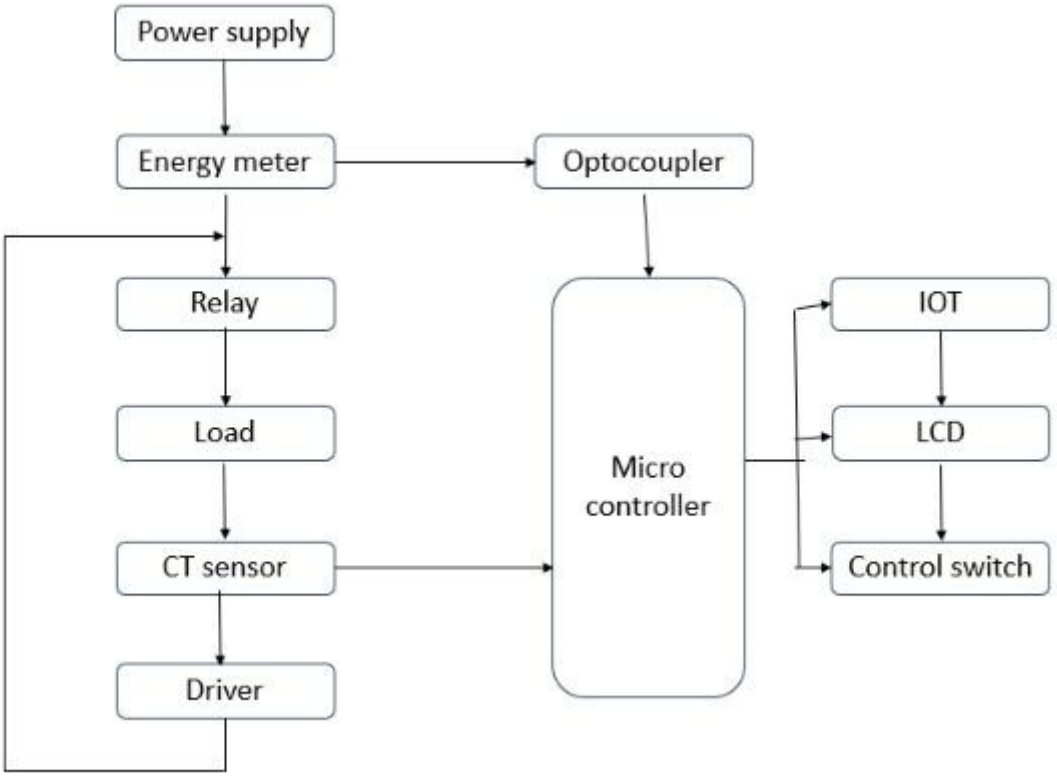


Fig:4.2.1 BLOCK DIAGRAM

The **Fig:4.2.1** has the following Explanation of each block:

1. Power Supply:

This block receives the 230V AC utility mains and steps it down to a safer 12V AC using an iron-core transformer. The stepped-down voltage is then rectified by a full-wave diode bridge into pulsating direct current. Filter capacitors smooth this output, which is subsequently regulated by LM7805 and

AMS1117 linear regulator ICs. It ultimately provides stable +5V and +3.3V DC rails to safely power the microcontrollers, sensors, and logic circuits.

2. Energy Meter:

This component is connected directly to the main incoming 230V AC supply to monitor the total power drawn by the household. It continuously measures instantaneous power and generates calibrated optical pulses proportional to the consumed kilowatt-hours. These optical flashes provide an accurate digital representation of the energy usage without requiring invasive electrical line modifications. By acting as the primary aggregate consumption tracker, it serves as the foundation for calculating tiered utility billing milestones.

3. Optocoupler:

This block bridges the high-voltage energy meter and the low-voltage microcontroller while providing vital galvanic isolation. It captures the optical calibration flashes from the meter using an internal infrared LED and an isolated phototransistor.

This conversion safely translates the high-voltage optical signals into clean 5V digital logic pulses for the microcontroller. It completely shields the sensitive processing circuitry from potentially destructive grid-borne surges, voltage spikes, and electrical noise.

4. Microcontroller:

Serving as the central processing unit, this block receives digital interrupt pulses from the optocoupler to calculate cumulative energy usage. It simultaneously reads analog voltage signals from the current sensors to determine the true RMS current of individual branch loads. The processor continuously evaluates these metrics against programmed dynamic tariff thresholds and safety overcurrent limits. Upon detecting a fault or tariff milestone, it instantly commands the relay drivers and formats telemetry data for wireless transmission.

5. Current Transformer (CT) Sensor:

The sensor is clamped non-invasively around the live phase wire of a specific household appliance branch circuit. It detects the alternating magnetic field generated by the active load and induces a proportional secondary current. This secondary signal is passed through a burden resistor and a DC biasing circuit to create a safe, measurable analog voltage. It enables granular, real-time current monitoring of individual devices without requiring any physical wire cutting.

6. Relay and Driver Module:

This block acts as the physical actuation mechanism to isolate hazardous or faulty appliances from the mains supply. It is controlled by a digital output from the microcontroller via a BC547 transistor switching stage. Under normal operating conditions, the relay is held energized to ensure uninterrupted power

delivery to the connected AC load. If an overcurrent anomaly is detected, the coil is rapidly de-energized to snap the contacts open and disconnect the specific appliance.

7. LCD Display:

The 16x2 character display acts as the localized user interface for immediate on-site monitoring of the energy management system. It is wired directly to the microcontroller to provide continuous, real-time readouts of critical electrical parameters. The screen actively displays the total consumed units, live individual branch currents, and system health status. This ensures that occupants have full visibility of their energy usage and localized fault warnings even during network or internet outages.

8. IoT Module:

This dedicated NodeMCU ESP8266 Wi-Fi transceiver handles all external wireless communication and cloud connectivity. It receives packaged serial telemetry data, including energy metrics and fault flags, directly from the main microcontroller via UART. Operating on an isolated 3.3V rail, it utilizes web protocols to push this real-time information to an external cloud dashboard. It ultimately facilitates remote user monitoring and dispatches automated push notifications for tariff alerts and electrical hazard warnings.

4.3 CIRCUIT DIAGRAM:

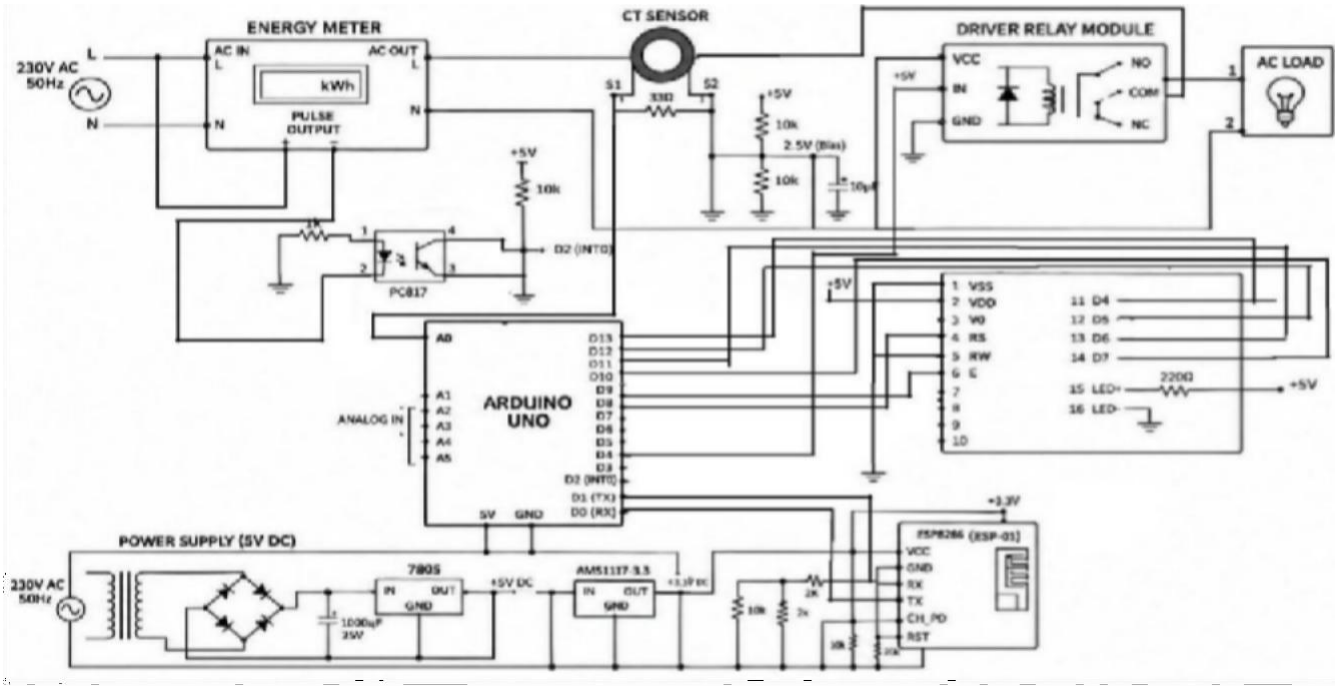


Fig:4.3.2 CIRCUIT DIAGRAM

The **Fig:4.3.2** The circuit requires stable, regulated direct current to power the sensitive electronic components accurately. A step-down transformer first reduces the incoming 230V alternating current mains to a safer 12V AC, which is then converted into pulsating DC by a full-wave diode bridge rectifier. To ensure smooth operation, a 1000µF filter capacitor removes the ripple, and an LM7805 linear voltage regulator locks the output to a steady +5V DC rail for the Arduino, relay, and LCD. Because the ESP8266 Wi-Fi module strictly operates on 3.3V and can be damaged by higher voltages, a secondary AMS1117-3.3

regulator safely steps down the 5V line to provide a dedicated 3.3V power source for the IoT components.

For electrical measurement, the 230V AC mains passes through a commercial single-phase energy meter, which outputs optical calibration pulses proportional to the consumed energy. To protect the microcontroller from grid-level voltage spikes, these pulses are routed through a PC817 optocoupler, which uses an internal phototransistor to pull the Arduino's hardware interrupt pin LOW, enabling safe pulse counting. Simultaneously, the AC load branch is monitored by a split-core Current Transformer (CT) clamped around the live phase wire. The induced secondary current from the CT sensor passes through a 33Ω burden resistor and a +2.5V DC biasing circuit, shifting the alternating waveform into a positive voltage range that the Arduino's analog pin can safely sample to calculate RMS current.

The Arduino Uno acts as the central logic controller, orchestrating the sensor data acquisition, protective actuation, and external communication. Under normal operating conditions, the Arduino outputs a digital signal to the driver relay module, keeping the normally open (NO) contacts closed to power the AC load. If an overcurrent fault or tariff cutoff is detected, the Arduino de-asserts this pin, opening the relay to isolate the specific appliance immediately. Concurrently, the Arduino updates a 16x2 LCD screen for localized real-time visibility and transmits the calculated energy metrics to the ESP8266 Wi-Fi module via a

voltage-divided UART serial connection, allowing the telemetry to be published to a remote cloud dashboard.

4.4 FLOW CHART

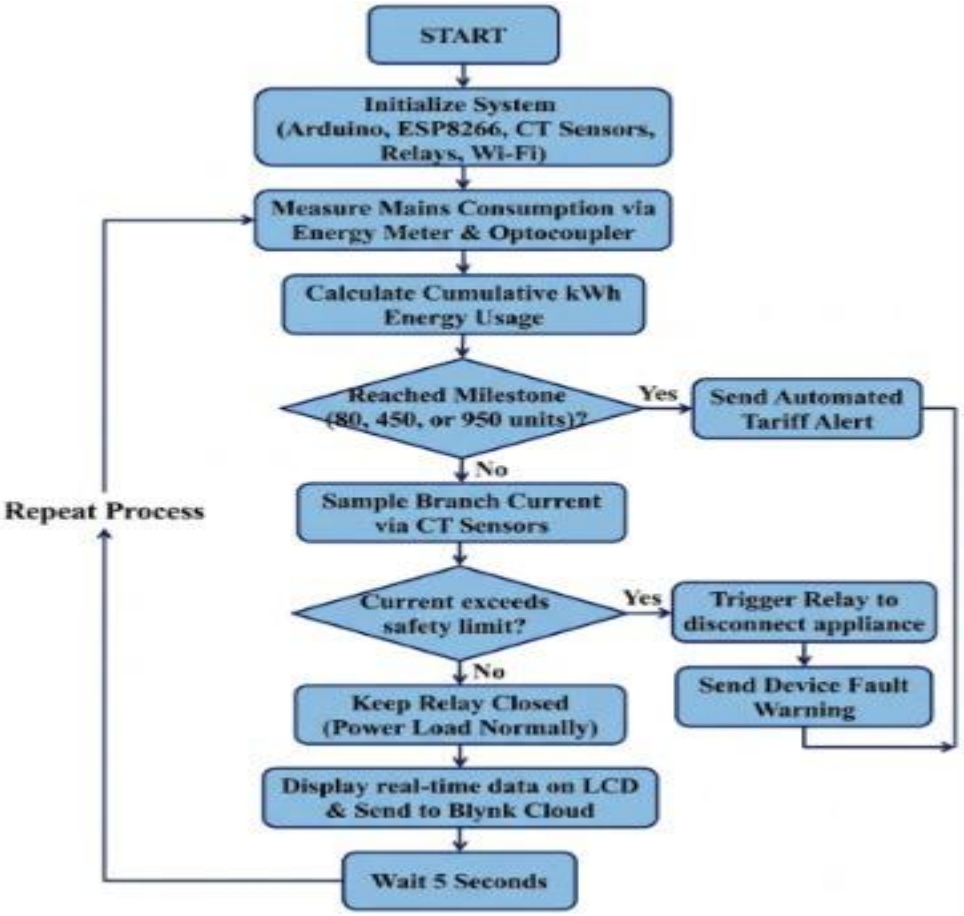


Fig:4.4.3 FLOW CHART

The operational flow of the proposed system begins with the initialization of the central microcontrollers, including the Arduino Uno and ESP8266, alongside the CT sensors, relays, and Wi-Fi modules. Once the system is active, it continuously measures the total mains electricity consumption by capturing the optical calibration pulses from the commercial single-phase energy meter. These

pulses are safely routed through a PC817 optocoupler to the Arduino, which processes the interrupts to calculate the cumulative kilowatt-hour (kWh) energy usage in real time.

Following the calculation of total energy consumption, the microcontroller evaluates this data against predefined dynamic tariff milestones. The system checks if the cumulative kWh has reached specific critical thresholds, specifically at 80, 450, or 950 units. If any of these milestones are crossed, the system immediately triggers an automated tariff alert. This alert is processed and transmitted to the user via the cloud to proactively warn them of impending higher billing slabs, allowing for timely adjustments in energy usage.

Concurrently with the main-line metering, the system executes branch-level appliance monitoring by sampling the live current through dedicated Current Transformer (CT) sensors. The Arduino reads these analog values and calculates the true current, which is then compared against a programmed safety limit. If the current exceeds this safe operating threshold—indicating a degrading component or severe overload—the system instantly triggers the protective relay to disconnect the malfunctioning appliance and generates a dedicated device fault warning.

If the branch current remains within normal, safe limits, the microcontroller maintains an active control signal to keep the relay closed, ensuring uninterrupted power delivery to the connected household loads.

Regardless of the fault status, the system continuously compiles all real-time electrical parameters and displays them locally on the 16x2 LCD screen. Simultaneously, this telemetry data is forwarded to the NodeMCU, which uploads it to the Blynk IoT cloud dashboard for remote monitoring. The system then waits for a brief 5-second sampling interval before repeating the entire continuous monitoring cycle.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 RESULTS

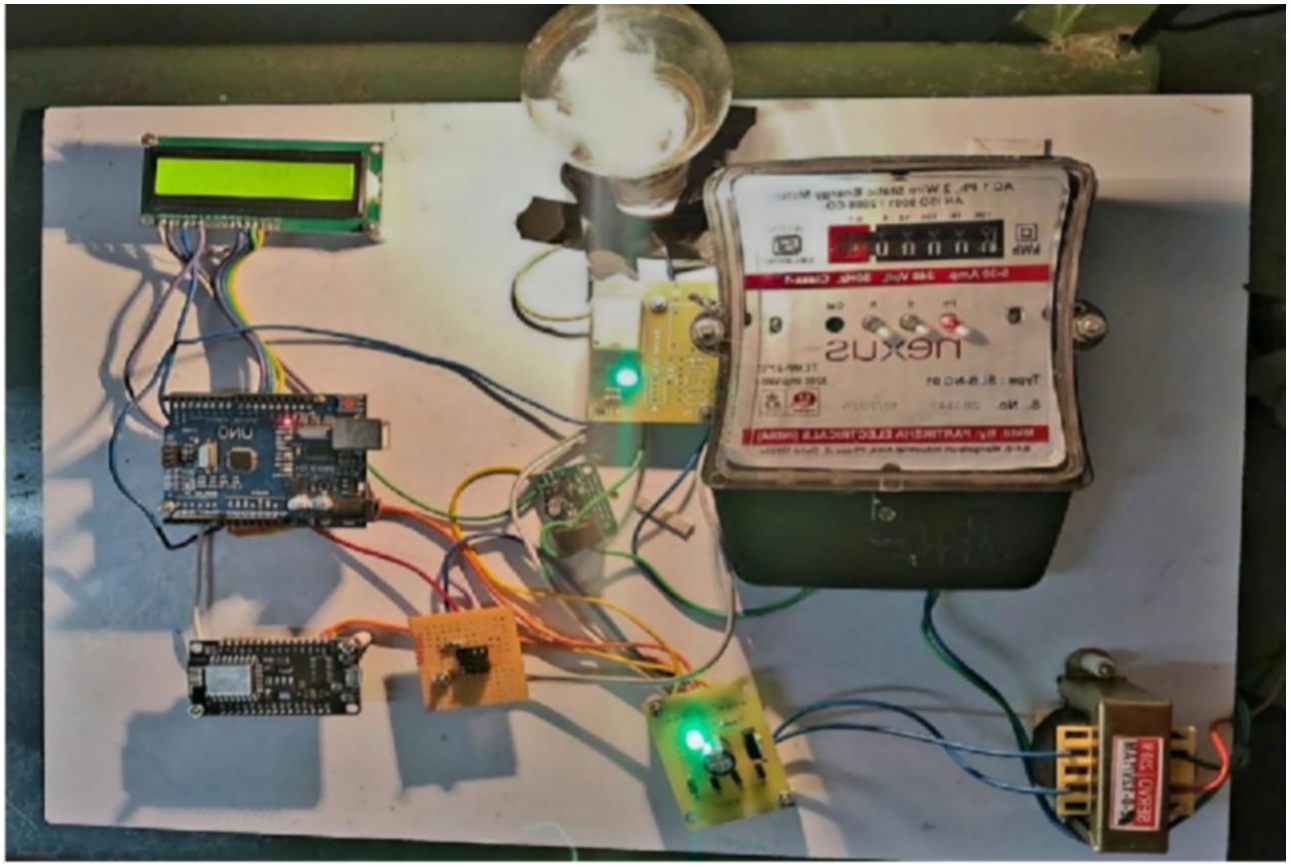


Fig:5.1.1 PROTOTYPE

The physical prototype integrates all computational, sensing, isolation, and actuation subsystems onto a single demonstration board, maintaining a clear physical separation between the high-voltage 230V AC power handling elements and the low-voltage DC logic circuits.The system is powered by a step-down transformer that reduces the mains supply to an isolated 12V AC, which is then rectified and smoothed by a dedicated power supply unit. An onboard LM7805

regulator ensures a ripple-free +5V DC rail for the main components, while a secondary regulator provides an isolated +3.3V DC line specifically to stabilize the Wi-Fi module.

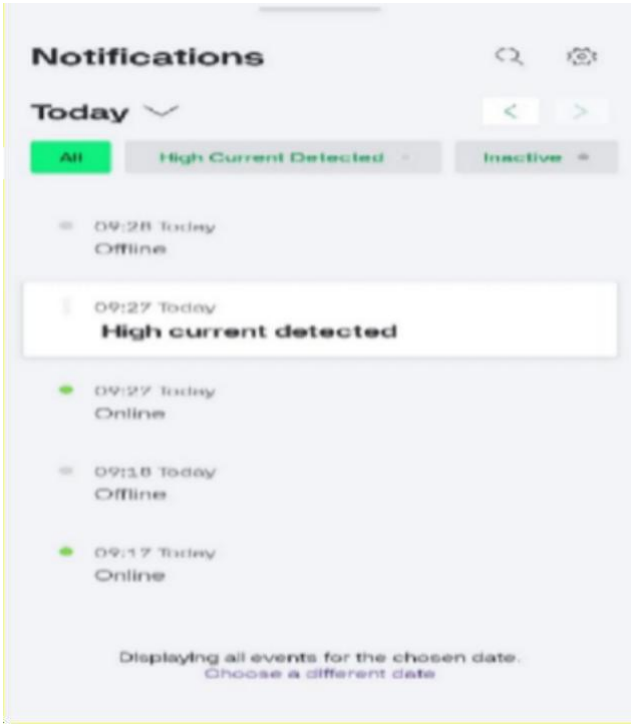


Fig:5.1.2 MOBILE NOTIFICATION

At the top of the demonstration board, a commercial Nexus static energy meter is mounted to measure cumulative household energy consumption directly from the grid. To protect the digital microcontrollers from electrical spikes, the meter's internal calibration pulse leads are connected to an isolated prototyping board containing a PC817 optocoupler. This optocoupler safely converts the internal optical flashes into clean 5V digital logic pulses, which are routed

directly into the hardware interrupt pin of the Arduino Uno for precise unit counting.

For branch-level monitoring, a dedicated toroidal Current Transformer (CT1270) sensor is situated in the center of the chassis to monitor the target appliance, represented in the prototype by an incandescent test bulb. The live phase wire originating from the energy meter passes directly through the aperture of this CT ring, allowing the sensor to output a proportional induced secondary signal into the Arduino's analog pin for continuous RMS current calculation. Adjacent to the load is a 5V electromechanical relay driver module, driven by a BC547 transistor, which the Arduino utilizes to physically break the circuit and safely isolate the load when an overcurrent anomaly is detected.

The Arduino Uno, positioned centrally on the board, acts as the primary data processing hub by running the core firmware routines required for intelligent energy management. It continuously captures the high-speed pulse interrupts to record consumed units, samples the analog current levels, and executes threshold evaluations for both multi-tier tariff alerts (at 80, 450, and 950 units) and overcurrent tripping. To provide immediate on-site visibility, a 16x2 LCD screen is mounted on the board, offering localized, real-time readouts of the units consumed, instantaneous current, and system fault messages.

The final system results are demonstrated through the successful wireless telemetry managed by the NodeMCU (ESP8266) Wi-Fi module, which receives

serial data packets from the Arduino over a dedicated UART link. This module seamlessly pushes the real-time energy data to the Blynk IoT dashboard, enabling comprehensive remote monitoring for the user. The practical output of this setup is validated by the system's ability to trigger automated mobile push notifications—such as timestamped "High current detected" alerts—proving its effectiveness in delivering immediate fault diagnostics and proactive tiered tariff warnings directly to a smartphone.

5.2 GRAPH ANALYSIS

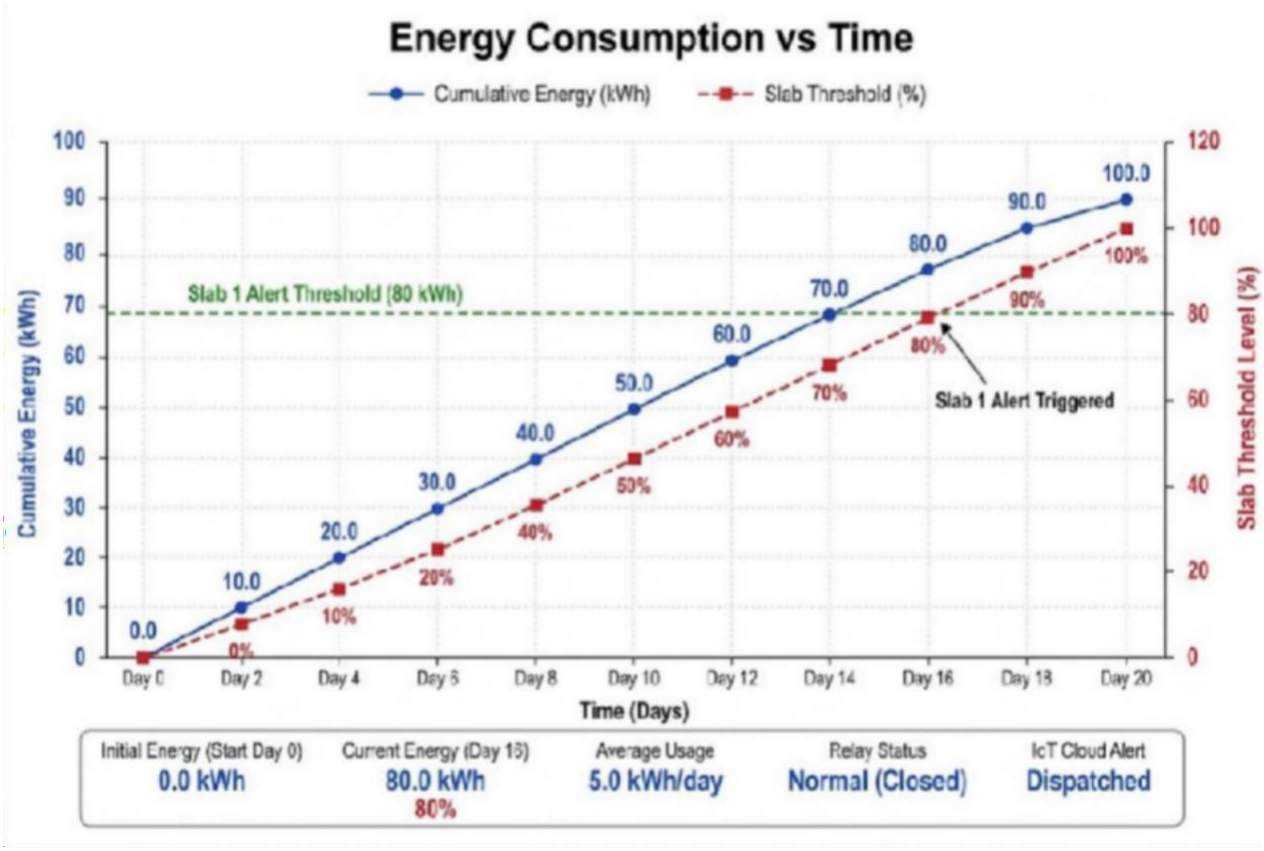


Fig:5.1.2 ENERGY CONSUMPTION GRAPH

The graphical analysis of the IoT-Based Intelligent Energy Management System illustrates the relationship between cumulative energy consumption and time over a continuous 20-day monitoring period. The primary vertical axis represents the total aggregate energy recorded by the commercial single-phase static energy meter, measured in kilowatt-hours (kWh), while the horizontal axis denotes the progression of time in days. A secondary vertical axis is utilized to track the normalized percentage progression toward the first critical billing threshold, providing a clear visual representation of how quickly the household is approaching the higher tariff slab.

Under typical domestic operating conditions, the system records a steady and linear accumulation of household power consumption. Starting from an initial baseline of 0.0 kWh on Day 0, the consumption trajectory climbs consistently to reach 100.0 kWh by Day 20. This steady climb reflects an average daily electrical usage of approximately 5.0 kWh per day. By capturing this continuous data stream, the microcontroller is able to map the exact trajectory of energy accumulation in real time, preventing sudden billing surprises at the end of the month.

The most critical functional feature demonstrated in the graph is the automated multi-tier tariff slab detection, highlighted by the horizontal threshold line set at 80 kWh. On Day 16, as the cumulative energy consumption strictly intersects this pre-programmed boundary, the system's internal threshold routine

is instantly triggered. In response, the microcontroller immediately updates the local display to warn the user and commands the NodeMCU ESP8266 to dispatch a real-time push notification via the IoT cloud dashboard. Because this milestone represents an informational tariff warning designed to encourage energy conservation rather than an electrical overcurrent fault, the system's relay module remains in its normal closed state, ensuring that the household continues to receive uninterrupted power

5.3 DISCUSSION

The integration of Internet of Things (IoT) architecture with real-time electrical instrumentation introduces significant technical advantages over traditional metering setups. By incorporating independent Current Transformer (CT) sensors on individual load branches, the system provides granular, real-time visibility into the exact Root Mean Square (RMS) current drawn by specific appliances. Unlike conventional installations where an overcurrent fault trips the main Miniature Circuit Breaker (MCB) and causes a total household blackout, this system uses dedicated electromechanical relays to isolate only the faulty load within milliseconds, leaving the rest of the house powered. Furthermore, the system ensures robust high-voltage protection by utilizing a PC817 optocoupler to provide galvanic isolation, effectively shielding the sensitive low-voltage microcontrollers from grid-borne surges and electrical noise.

From an economic and consumer-centric perspective, the proposed system actively mitigates bill shock by replacing reactive monthly billing with proactive dynamic tariff alerts. The system continuously evaluates cumulative kilowatt-hour pulses against tiered utility rate milestones—specifically at 80, 450, and 950 units—and dispatches automated IoT notifications before higher, more expensive billing brackets are crossed. Additionally, the system provides crucial protection against hidden appliance degradation. By continuously monitoring for aging appliances that draw persistent excess current, it prevents undetected energy waste, mitigates potential electrical fire hazards, and allows users to remotely disconnect inactive devices to eliminate parasitic standby power losses.

Practically, the system is designed for cost-effective scalability and non-invasive retrofitting into existing residential infrastructures. Branch current monitoring is achieved by simply clamping split-core CT sensors over existing insulated phase conductors, and main-line metering relies on capturing optical flashes, completely avoiding dangerous high-voltage wire splicing or modifications to utility-sealed enclosures. The architecture also boasts a resilient, dual-layer interface, providing immediate on-site visibility via a 16x2 LCD screen during internet outages, while the NodeMCU ESP8266 simultaneously handles remote cloud telemetry via dashboards like Blynk or Firebase.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The design and implementation of the "IoT-Based Intelligent Energy Management and Abnormal Load Monitoring System" successfully address the critical shortcomings of conventional whole-house electricity meters. By combining total main-line power measurement with dedicated branch-level current sensing, the system achieves a balanced framework for both financial management and electrical safety. The integration of the commercial static energy meter with an optically isolated PC817 optocoupler allows the Arduino Uno to accurately track cumulative consumption without exposure to high-voltage transients. Additionally, the dynamic slab warning mechanism successfully triggers proactive alerts at critical tariff milestones—specifically at 80, 450, and 950 units—empowering consumers to regulate their energy usage before incurring expensive higher-slab charges.

Furthermore, branch-level monitoring via the CT1270 current sensor provides continuous visibility into individual appliance health. When abnormal current spikes or overload faults are detected, the system automatically triggers a 5V relay module to isolate the malfunctioning appliance within milliseconds, mitigating hidden energy waste and preventing electrical hazards without shutting

down power to the rest of the household. Telemetry streaming via the NodeMCU ESP8266 ensures continuous local and cloud visibility, establishing an effective prototype for modern smart energy management.

6.2 FUTURE SCOPE

The Future enhancements to the system could involve integrating machine learning algorithms to enable predictive maintenance, allowing the cloud dashboard to analyze long-term current consumption patterns and forecast appliance failures before they occur. The hardware architecture can be scaled from single-phase to three-phase power systems for industrial applications by incorporating multiple current and voltage sensors to monitor phase imbalances, harmonics, and power factor anomalies. Furthermore, the system could be expanded to interface with renewable energy sources—such as residential solar inverters—to intelligently switch household loads between grid power and battery storage based on real-time dynamic tariff rates. Finally, interoperability can be improved by integrating the IoT framework with mainstream smart home ecosystems for voice-activated load control, alongside the development of a dedicated native mobile application to provide users with advanced historical energy auditing and automated budget forecasting.

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